ZYVORAI LABS · CUSTOMER DOCUMENTATION

Zeus OS

Admin Basics

Visual day-2 control plane for KubeVirt and Kubernetes

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Platform-facing configuration for Zeus OS. Operators using the UI day-to-day can skip this unless asked to verify URL, TLS, or auth.

For the full env-var and production checklist, see Handbook — Administration & Configuration.

Ports customers care about

Port	Service	Typical use
5050	`zeus-os-web`	Direct / local dashboard + API
30050	NodePort (common deploy)	Cluster-exposed HTTPS dashboard
8081	Optional console-proxy	Sidecar console proxy
9191 / 9443	PacketWolf	Network intelligence (proxied)
9090	Prometheus	Metrics source (cluster)

Zeus itself listens on **5050** (plus optional console proxy). Give customers the **ingress or NodePort URL**, not internal ClusterIP addresses.

Fastest production-style deploy

```
# On a machine with kubeconfig + KubeVirt already installed
./scripts/deploy-remote.sh <HOST> <SSH_USER> --k3s
# → https://<HOST>:30050
```

Air-gapped: AIR_GAPPED_INSTALL.md. Helm profiles: HELM_VALUES_PROFILES.md.

From a binary bundle (no git clone, no build tools)

If you received a `zeus-os-<version>-linux-amd64.tar.gz` bundle (built via `scripts/package-binary-remote.sh`), it ships the Helm chart plus a script that builds a runtime image from the bundle's own binaries — no source tree, no cargo/npm:

```
tar xzf zeus-os-*-linux-amd64.tar.gz && cd zeus-os-*-linux-amd64
./install-cluster.sh                                        # KubeVirt/CDI/Cilium prereqs
./install-to-kubernetes.sh --registry <your-registry>/zeus-os --push
./apply-cluster-network.sh
```

`install-to-kubernetes.sh --help` shows namespace/release/dry-run options, and a local single-node cluster (kind/k3s/minikube) sharing the build daemon's image store can skip `--push` entirely. Note: an image built this way lacks the `helm` CLI the full-source `web/Dockerfile` bundles — the in-app /helm repo/install UI won't work; use `web/Dockerfile` instead if you need it. Browser-VNC assets and GuestKit agent binaries are included.

Authentication (what to enable)

Resolution order on the server:

1. `ZEUS_OS_NO_AUTH=1` — lab only
2. `ZEUS_OS_API_KEY` — static bearer for API
3. `ZEUS_OS_AUTH=local|ldap|saml|oidc` — interactive login

Enterprise detail: ENTERPRISE_AUTH_TENANCY.md.

When `ldap` / `saml` / `oidc` is set, the local admin account stays available as a break-glass fallback automatically — the login screen shows an explicit toggle to pick between the directory/ SSO method and the local admin account, rather than requiring the operator to know a hidden URL.

Tell operators:

- Dashboard URL + auth method
- Whether MFA / SSO is required
- That RDP/SSH use **guest** credentials

Local admin password

The built-in local admin account always uses a password you choose at install time. The Helm chart refuses to install without one, and the binary bundle generates a random one.

How it is set differs by install path:

Install path	Behavior
Helm chart	**Refuses to install** unless `config.localAuth.adminPassword` or `config.localAuth.adminPasswordSecretRef` is set (`config.localAuth.allowDefaultPassword=true` exists for throwaway dev clusters only)
Binary bundle (`scripts/package-binary-remote.sh`)	Generates a **random** strong password automatically unless `ZEUS_OS_LOCAL_ADMIN_PASSWORD` is set in the bundle's env file — does **not** fall back to a fixed default

Set your own password at install time (Helm):

```
helm upgrade --install zeus-os ./chart --set config.localAuth.adminPassword="a-real-password"
# or reference an existing Secret instead of inlining the password:
# --set config.localAuth.adminPasswordSecretRef.name=zeus-os-local-auth
```

TLS (required for browser VNC)

The Helm chart serves **HTTPS by default**. With no TLS settings it generates a self-signed certificate (valid ten years, with the Service names and `localhost` as names) and reuses it on every `helm upgrade`, so the fingerprint you accepted in the browser does not change. Browsers warn about a self-signed certificate; bring your own to remove the warning:

You want	Set
Your own certificate	`kubectl create secret tls zeus-os-tls --cert=tls.crt --key=tls.key`, then `--set tls.secretName=zeus-os-tls`
Extra names in the generated certificate	`--set 'tls.hosts[0]=zeus.example.com'`
TLS terminated at an Ingress	`--set ingress.enabled=true` (the pod stays on HTTP unless `tls.secretName` is set)
Plain HTTP (only behind something that terminates TLS)	`--set tls.selfSigned=false`

Without the chart, set the certificate directly:

```
export ZEUS_OS_TLS_CERT=/path/to/cert.pem
export ZEUS_OS_TLS_KEY=/path/to/key.pem
```

Without a secure context, in-browser VNC will not work; serial console still can.

Multi-tenancy

Namespaces may be labeled `zeus-os.io/tenant=<id>`. List APIs filter by tenant for scoped users. Configure tenants under **Tenancy** (`/tenancy`) and Identity Center.

What to hand customers after install

1. Correct HTTPS URL (and the port that is *not* GuestKit/Machina).
2. Login instructions for your auth mode.
3. Link to this customer package: `docs/customer/README.md` (or the PDF set).
4. Optional: Feature Guide PDF for capability overview.

Health checks

```
curl -sk -o /dev/null -w "%{http_code}\n" https://<host>:30050/api/v1/health
kubectl -n zeus-os get pods,svc   # namespace may differ per chart
```

Related

- Getting Started — operator path after URL works
- Enterprise SSO / OIDC — Keycloak and generic OIDC
- CUSTOMER_ACCESS_RUNBOOK.md — example access notes
- Handbook README